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Wealth based inequality in the prevalence of teenage pregnancy and its correlates: Evidence from National Family Health Survey, 2019-21

Saurabh Singh[1] and Neha Shri[1*]

Abstract

Background The inclusion of adolescent fertility in the Sustainable Development Goals demonstrates the widespread recognition of the possible health hazards as well as social and economic disadvantages encountered by young women and their infants. Moreover, adolescents are in a precarious situation due to their socioeconomic vulnerability coupled with the responsibility of motherhood at an early age. The present study has been conducted to examine factors associated with adolescent fertility in India and to understand the socioeconomic inequality in teenage pregnancy utilizing data from the National Family Health Survey-5 (NFHS-5).

Methods Binary logistic regression and marginal effects are applied to determine the association between teenage childbearing and other independent variables. The socioeconomic inequality in teenage pregnancy is assessed by Errygers decomposition.

Results Around one-fifth of the respondents aged 19 years reported being pregnant in their teenage (21%). The risk of teen pregnancy was negatively associated with increasing educational attainment, improving wealth index, and working status. The risk of teenage pregnancy was 46% lower among individuals with family planning mass media exposure (OR: 0.54***0.4–0.75). Marginal effects showed that with secondary and higher educational attainment, the probability of having teenage pregnancy reduced by three and eight percent points in comparison to women without education. Results from Erregyers decomposition indicated that variables such as rural place of residence (47%), attempt to delay pregnancy (25%), no exposure to mass media related to family planning (13%), incorrect knowledge of ovulatory cycle (4%) were positive contributors to inequality attributable to wealth quintile in teenage pregnancy.

Conclusions Promoting formal, institutional sources of knowledge regarding sexual and reproductive health, availability of contraception and their effective usage can enable adolescents to understand the negative repercussions of adolescent pregnancy. Moreover, work participation, higher schooling and access to family planning can also positively contribute to the reduction of teenage pregnancies.

Keywords Teenage pregnancy, Wealth inequality, Adolescents, Erregyers decomposition, India

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Background

Around the world, an estimated 21 million girls aged 15–19 years become pregnant annually in developing countries and some 12 million births take place [1]. Pregnancy and delivery complications are among the major causes of death for females between the ages of 15 and 19 worldwide [2]. Additionally, teenage mothers under the age of 15 have the highest risk of maternal mortality, while adolescent girls between the ages of 10 and 19 have the highest risk of pregnancy and childbirth complications (compared to women aged 20–24) [3]. The inclusion of adolescent fertility as a crucial indicator for reproductive health in the Sustainable Development Goals demonstrates the widespread recognition of the possible health hazards as well as social and economic disadvantages encountered by these young women and their infants [4]. The consequences of adolescent pregnancy and childbirth are enormous in terms of increased morbidity and mortality [5] for the mother and child as well as long-term health care costs [6]. An adolescent pregnancy threatens girls' health, restricts their socioeconomic opportunities and has negative repercussions at the individual, community and society levels. On the other side, children born to these adolescent mothers are likely to have poorer behavioral, educational and health outcomes throughout their life course [7]. These unintended pregnancies are more likely to be aborted [8] with abortions performed by unskilled providers lacking necessary skills and under unhygienic conditions [9].

Majority of the Indian women become sexually active and start childbearing between the ages of 15–19 years [1]. Additionally, around 3.4 million adolescent women want to avoid pregnancy however only about 29% are using modern contraceptive methods in the country [1]. Almost two-thirds (63%) of adolescent pregnancies are unintended accounting for nine out of every ten unintended pregnancies taking place in the absence of modern contraception [1]. The unmet need for contraception among adolescents is comparatively higher for those from rural areas, those who haven't begun childbearing, and those younger than 18. Half of these adolescent pregnancies end in abortion of which an estimated 78% are unsafe (abortions performed in unsafe environments and/or performed by a person lacking the necessary skills) [1]. Although abortion is legal under the Medical Termination of Pregnancy (MTP) Act, 1971, access to safe abortion services remains limited, particularly for disadvantaged populations, contributing to the high burden of unsafe abortions. To worsen the scenario, despite a substantial decline in adolescent fertility in India [1], around 7% of women aged 15–19 years have begun childbearing [10].

Research indicates that most often these adolescent mothers are poorer and less educated [7]. Further, a major proportion of these pregnancies are often unwanted and are driven by a lack of knowledge, limited access to contraceptives, fewer bodily rights, and a lack of control over contraceptive methods [9]. Given the high prevalence of unintended pregnancies and unsafe abortions taking place in the country, various health policies are in action to improve access to contraception and foster a supportive environment for addressing the specific needs of adolescents. However, these young women are in a precarious situation due to their socioeconomic vulnerability coupled with the responsibility of motherhood at an early age. An empirical study of the variables affecting teenage girls' pregnancy rates would broaden our understanding of the subject and enable medical professionals the tools they need to deliver better health care to this population.

Addressing teenage pregnancy is critical to achieving global development targets, particularly the Sustainable Development Goal (SDG). SDG 3 aims to ensure healthy lives and promote well-being for all at all ages, including reducing maternal mortality and improving adolescent health. SDG 5 emphasizes the need to eliminate all harmful practices such as early and forced child marriage and to ensure universal access to sexual and reproductive health and reproductive rights. Analysing the factors that contribute to inequality is essential for understanding the structural and social determinants that shape health outcomes, such as teenage pregnancy. Wealth-based inequality reflects differences in access to resources like education, healthcare, nutrition, and information, which directly influence adolescent reproductive behaviour and choices. By analysing wealth-based inequalities in teenage pregnancy, this study contributes evidence needed for policy formulation that promotes adolescent health and gender equality. Thus, the purpose of this study is to examine factors associated with adolescent fertility and wealth based inequality in the prevalence of teenage pregnancy in India.

Methods

Data Source

This study utilizes data from the most recent round of the National Family Health Survey (i.e., NFHS-5) conducted in 2019–2021. NFHS-5 was conducted under the direction of the Ministry of Health and Family Welfare, Government of India, with the International Institute for Population Sciences, Mumbai, as the nodal agency. It is a nationally representative, cross-sectional survey that provides information on fertility, infant and child morbidity & mortality, family planning, and other important aspects of women-child health and nutrition. The survey includes 476,561 households from rural and 160,318 households from urban areas. More information on the

sample design and survey details can be obtained from the NFHS-5 Report [10].

This study uses Individual (women's) files. Based on the outcome variable of interest, women in the age group 15–19 years have been selected for the present study. A total of 122,477 women were eligible for the study. The sample focused on currently pregnant women and/or those with at least one child or ever had a pregnancy.

Variables

Outcome Variable

Teenage childbearing is the main outcome variable of this study. It can be defined as the teenage girls (15–19 years in this study) either currently pregnant or ever pregnant irrespective of their pregnancy outcome. A dummy variable was created and coded as 1 if the teenager was pregnant at the time of the survey or experienced pregnancy and 0 otherwise.

Independent Variables

Based on previous pieces of literature, the following variables are used as predictors of teenage pregnancy and have been classified as individual and community-level factors:

Individual Level Variables

The age of respondents was taken in single completed years at the time of the survey. Current marital status was recoded into two categories (1 if currently married and 0 = otherwise). Religion has subcategories of 'Hindu', 'Muslim', and 'Others' based on their religious beliefs. Caste/Ethnicity was recoded into four categories- 'SCs' (Scheduled Castes), 'STs' (Scheduled Tribes), 'OBCs' (Other Backward Castes), and 'Others'. Educational attainment of women was categorised into 'without any formal education', 'primary education', 'secondary education', and 'higher education'. Wealth index was categorized into five quintiles (poorest to richest) derived based on assets and amenities in the household. Employment status was recoded as 'currently working' and 'currently not working'. Watching television (TV), listening to the radio and reading the newspaper at least once a week were merged and a dichotomized variable was generated as "having mass media exposure" and "not having mass media exposure." The household head's age and sex were categorized as less than 30 years, 30–44 years, 45–59 years and 60 or above & male and female respectively. The respondents were asked if they ever tried to avoid or delay pregnancy, knowledge of modern methods of contraception and their responses were recoded as yes or no. Based on the response to the question "Have you ever heard family planning on either TV or radio or magazine/newspaper" the variable 'exposure to family planning mass media' was coded as (none, one, two, three). Information on knowledge about the ovulatory cycle was recoded as incorrect, correct and don't know.

Community Level Variables

The place of residence was dichotomized into urban and rural. The region was the geographical location where the population lived. The states and Union Territories of India have been categorized into six regions, namely North, Central, East, Northeast, West, and South.

Statistical analysis

Descriptive Statistics like frequency and percentage have been used to describe the distribution of population. Further, bivariate analysis was computed to examine the prevalence of teenage pregnancy by various socio-economic and demographic characteristics. Also, Pearson's chi-square test was used to identify the significance of such associations. Since the chi-square test doesn't indicate the strength (effect size) of a statistically significant relationship, Cramer's V was used. It is calculated by dividing chi-square values by sample size and taking the square root of this value. The outcome variable of the study is dichotomous so binary logistic regression was applied to determine the association between teenage childbearing and other independent variables in terms of crude odds ratio with a 95% confidence interval. Also, the marginal effect and its 95% confidence interval were calculated. The study found no evidence of multicollinearity among the independent variables using the Variance Inflation Factor (VIF) (mean VIF = 1.14).

To measure the socioeconomic inequality in teenage pregnancy, this study used a concentration index. The concentration curve is a plot of the cumulative share of teen pregnancy against the share of the population in ascending order of the wealth index. The magnitude of the concentration index is twice the area between the concentration curve and the line of perfect equality (45-degree line). If the concentration curve lies above (or below) the line of perfect equality, it suggests that teenage pregnancy is concentrated among the poor (or rich).

The value of the concentration index was calculated using formula- $CI = \frac{2}{\mu}\mathrm{cov}\,(h,\, r)$

Where CI indicates the concentration index, is the mean of teenage pregnancy, h is the teenage pregnancy and r is the cumulative percentage that total adolescents represent over the total population after ranking teenage pregnancy by the wealth index.

The concentration index value ranges from – 1 to 1 for continuous outcome variables. Since the outcome variable of this study is binary, the value of the concentration index depends on μ. For the prevalence of teenage pregnancy, an Erreygers CI (ECI) [11], a modified version

of the normal CI is used to measure the socio-economic inequality which can be defined as

$$ECI = 4 * \mu * CI\ (y)$$

.

Where CI(y) is the generalized CI and μ is the mean teenage pregnancy. This normalized value of the concentration index is used to quantify socioeconomic inequalities in teen pregnancy. Data analysis used appropriate weights to ensure the representation of the sample at the regional and national levels. STATA 16 was used for analysis.

Results

Table 1 presents the distribution of the sample by various socio-economic and demographic characteristics. The distribution of respondents did not vary much by single-year age. The majority of the respondents had a secondary level of education (83%). Around one-tenth of the respondents reported working (11%). Around a quarter of the study sample resided in urban areas (28%) and a little more than half of the sample had access to mass media (57%). The majority of the household heads were male and aged 45–59 years. Further, 95% of the women knew modern methods of contraception however only 7% of the young women tried to delay/avoid pregnancies. Around two-fifths of the women did not hear of family planning through mass media sources while only 11% had correct knowledge of the ovulatory cycle.

The prevalence of teen pregnancy was found to be around 7% (Table 1). Around one-fifth of the respondents aged 19 years reported being pregnant in their teenage (21%). A higher proportion of respondents without any formal schooling (19%), "poorest" households (11%), not working (7%) and residing in rural areas (8%) became pregnant in their teenage. The prevalence of teen pregnancy was higher among respondents from eastern India (13%) and those without any mass media exposure (9%). Further, teenage pregnancy was higher among those with access to family planning mass media (9%), modern contraception knowledge (8%) and correct knowledge of the ovulatory cycle (15%). Additionally, 50% of women who tried to delay/avoid pregnancy became pregnant. Results from the chi-square test indicated that respondent's age, caste, education, wealth index, religion, place of residence, region and mass media exposure were significantly associated with teen pregnancy ($p < 0.001$). Moreover, results from Cramer's V test showed a very strong association between teenage pregnancy and respondents age (Cramer's V: 0.29), tried to delay pregnancy (Cramer's V: 0.44), age of household head (Cramer's V: 0.24) and showed a moderate association between teen pregnancy and adolescent's educational status, geographical region & knowledge of ovulatory cycle(Cramer's V: 0.12).

Results obtained from logistic regression analysis are presented in Table 2. It was found that the odds of being pregnant in their teen ages increased with an increase in respondents' age. The risk of teen pregnancy was negatively associated with increasing educational attainment and improving wealth index. Further, the odds of teen pregnancy were 32% times lower among those who reported working (OR: 0.68*** CI: 0.54–0.86) and 1.6 times higher among rural residents (OR:1.59***CI:1.26–2.01) than non-working individuals and urban residents. The risk of teenage pregnancy was 46% times lower among individuals with three sources of family planning mass media exposure (OR: 0.54***0.4–0.75). Surprisingly, the risk of teenage pregnancy was higher among those with correct knowledge of the ovulatory cycle, knowledge of modern contraception methods, who tried to delay pregnancy and household heads aged 60 and above.

Results obtained from the average marginal effects are shown in Table 2. It was found that with the increase in the age, i.e. from 15 to 19, there was a subsequent increase in the proportion of teenage mothers. Girls aged 19 years were 14% point more likely to become mothers than girls aged 15 years. Moreover, with secondary and higher educational attainment, the probability of having teenage pregnancy reduces by three and eight percent points in comparison to women without education. Additionally, the probability of having a pregnancy in teenage was reduced when we moved from the poorest to the rich wealth quintile, non-working to working, and with exposure to mass media (overall mass media and family planning mass media). Further, when the place of residence changes from urban to rural, the change in probability increases by 2.1 points and is significant. Similarly, the average marginal change in the risk of teenage pregnancy increased by 18% & 3% points when they tried to delay pregnancy and had knowledge of modern contraception respectively.

Results from Erregyers decomposition indicated that variables such as rural place of residence (47%), tried to delay pregnancy (25%), no exposure to mass media related to family planning (13%), incorrect knowledge of ovulatory cycle (4%) were positive contributors to inequality attributable to wealth quintile in teenage pregnancy. However, factors such as caste, working status, Central, Western and Southern regions, female household sex, knowledge of modern contraceptive methods, family planning mass media exposure, and correct knowledge of the ovulatory cycle contributed to propoor inequality (Table 3).

The concentration curve plot for teenage pregnancy lies well above the line of equality indicating that the

Table 1 Descriptive characteristics of the study population and proportion of adolescents (aged 15–19 years) who ever got pregnant or are currently pregnant by various socio-demographic characteristics, NFHS-5, 2019-21

Variables		Freq.	Percent	Teen pregnancy (in %)	*p*-value	Cramer's V
Respondent's age (in years)	15	25,005	20.16	0.28	< 0.001	0.29
	16	24,714	20.07	1.22		
	17	23,580	19.31	3.58		
	18	26,629	21.77	10.42		
	19	22,549	18.7	21.23		
Caste	SC	25,116	23.01	7.82	< 0.001	0.02
	ST	22,563	9.54	9.26		
	OBC	48,100	43.87	6.2		
	Others	26,698	23.58	7.76		
Education	No education	5,398	4.35	19.18	< 0.001	0.12
	Primary	6,435	5.19	13.36		
	Secondary	102,869	83.15	6.62		
	Higher	7,775	7.3	2.71		
Wealth index	Poorest	29,426	22.45	10.5	< 0.001	0.09
	Poorer	30,237	22.81	8.52		
	Middle	25,747	20.92	7.32		
	Richer	21,085	18.6	5.5		
	Richest	15,982	15.22	2.45		
Working status	No	16,106	88.89	7.35	< 0.10	0.01
	Yes	2,130	11.11	6.22		
Religion	Hindu	91,767	79.83	6.99	< 0.001	0.02
	Muslim	17,558	15.84	8.84		
	Others	13,152	4.33	5.78		
Residence	Urban	26,928	28.19	4.18	< 0.001	0.06
	Rural	95,549	71.81	8.43		
State	North	24,251	14.03	3.7	< 0.001	0.12
	Central	33,651	29.29	3.68		
	East	22,444	25.08	12.72		
	North-east	16,129	3.48	11.99		
	West	10,965	12.06	7.28		
	South	15,037	16.05	7.16		
Mass media exposure	No	54,706	43.1	8.84	< 0.001	0.06
	Yes	67,771	56.9	6.02		
Sex of household head	Male	102,979	83.98	7.28	>0.10	0.01
	Female	19,498	16.02	6.97		
Age of household head	< 30 years	4,330	3.47	39.66	< 0.001	0.24
	30–44 years	38,230	31.21	3.8		
	45–59 years	58,151	47.42	6.8		
	60 and above	21,766	17.9	8.06		
Tried to delay pregnancy	No	11,504	93.45	4.17	< 0.001	0.44
	Yes	7,433	6.55	50.94		
Knowledge of any method	No	5,769	4.71	1.26	< 0.001	0.05
	Yes	116,708	95.2	7.53		
Family planning mass media exposure	No exposure	46,930	37.38	8.65	< 0.001	0.05
	One mass media	32,644	26.72	8.04		
	Two mass media	31,960	26.87	5.2		
	Three mass media	10,939	9.03	5.02		
Knowledge of ovulatory cycle	Incorrect	73,632	60.51	8.04	< 0.001	0.13
	Correct	14,123	11.06	14.78		
	Don't know	34,722	28.43	2.58		
Total		122,477		7.23		

Table 2 Odds ratios of being pregnant and average marginal effects indicating a change in probability of being pregnant among adolescents (aged 15–19 years) who ever got pregnant by various socio-demographic characteristics, NFHS-5, 2019-21

Variables		Odds Ratio	Margins - delta method (dy/dx)
Respondent's age (in years)	15®		
	16	6.51***(2.54–16.64)	0.014***(0.009,0.020)
	17	17.44***(7.06–43.08)	0.035***(0.028,0.040)
	18	46.69***(19.17–113.7)	0.076***(0.069,0.080)
	19	103.52***(42.54-251.89)	0.135***(0.124,0.150)
Caste	SC®		
	ST	0.76**(0.59–0.98)	−0.011**(−0.020,0.000)
	OBC	0.87 (0.71–1.07)	−0.005 (−0.014,0.000)
	Others	0.95 (0.74–1.23)	0.002 (−0.012,0.010)
Education	No education®		
	Primary	1.03 (0.72–1.45)	0.001 (−0.019,0.020)
	Secondary	0.52***(0.4–0.68)	−0.032***(−0.047,−0.020)
	Higher	0.09***(0.05–0.15)	−0.077***(−0.093,−0.060)
Wealth index	Poorest®		
	Poorer	0.94 (0.77–1.15)	0.003 (−0.011,0.010)
	Middle	0.85 (0.67–1.08)	−0.006 (−0.016,0.000)
	Richer	0.77*(0.58–1.03)	−0.010*(−0.021,0.000)
	Richest	0.73 (0.5–1.07)	−0.012*(−0.026,0.000)
Working status	No®		
	Yes	0.68***(0.54–0.86)	−0.014***(−0.022,−0.010)
Religion	Hindu®		
	Muslim	0.97 (0.77–1.21)	0.001 (−0.010,0.010)
	Others	0.79 (0.58–1.08)	−0.009 (−0.019,0.000)
Residence	Urban®		
	Rural	1.59***(1.26–2.01)	0.017***(0.009–0.02)
State	North®		
	Central	0.84 (0.64–1.12)	0.005 (−0.013,0.000)
	East	2.31***(1.76–3.01)	0.032***(0.022,0.040)
	North-east	2.48***(1.83–3.38)	0.036***(0.023,0.050)
	West	2***(1.45–2.76)	0.026***(0.013,0.04)
	South	2.77***(2.02–3.79)	0.041***(0.028,0.05)
Mass media exposure	No		
	Yes	0.83**(0.7–0.99)	−0.007**(−0.014,0.00)
Sex of household head	Male		
	Female	0.8**(0.65–0.98)	−0.008**(−0.016,0.00)
Age of household head	< 30 years		
	30–44 years	0.24***(0.18–0.32)	−0.079***(−0.098,−0.060)
	45–59 years	0.23***(0.18–0.29)	−0.081***(−0.099,−0.06)
	60 and above	0.3***(0.22–0.39)	−0.071***(−0.09,-0.05)
Tried to delay pregnancy	No		
	Yes	11.79***(9.99–13.91)	0.181***(0.163,0.200)
Knowledge of any method	No		
	Yes	2.39**(1.24–4.63)	0.027***(0.011,0.040)
Family planning mass media exposure	No exposure		
	One mass media	0.75***(0.62–0.91)	−0.012***(−0.019, 0.00)
	Two mass media	0.62***(0.51–0.77)	−0.018***(−0.0026,−0.01)
	Three mass media	0.54***(0.4–0.75)	−0.023***(−0.034,−0.01)
Knowledge of ovulatory cycle	Incorrect		
	Correct	1.43***(1.18–1.74)	0.016***(0.007,0.030)
	Don't know	0.52***(0.42–0.64)	−0.023***(−0.030,-0.020)

***Significant 99% confidence level, **significant at 95% confidence level, *significant at 90% confidence level

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prevalence of teenage pregnancy is higher among the poor (Fig. 1). This indicated that teenage pregnancy is more prevalent among adolescents from poorer households compared to their richer counterparts. The negative value of the concentration index (−0.080) confirms this pro-poor concentration teenage pregnancy disproportionately affects economically disadvantaged groups.

Table 3 Results obtained from errygers decomposition showing the wealth-based inequality in the prevalence of teenage pregnancy among adolescents (aged 15–19 years) who ever got pregnant by various socio-demographic characteristics, NFHS-5, 2019-21

		Elasticity	Concentration index	Contribution	Percentage contribution
Respondent's Age (in years)	15®				
	16	0.033	−0.004	0.000	0.15
	17	0.067	0.005	0.000	−0.46
	18	0.108	−0.007	−0.001	0.95
	19	0.118	0.034	0.004	−5.01
Caste	SC®				
	ST	0.000	−0.138	0.000	−0.03
	OBC	0.001	0.068	0.000	−0.07
	Others	0.007	0.171	0.001	−1.51
Education	No Education®				
	Primary	0.000	−0.075	0.000	0.02
	Secondary	−0.077	0.052	−0.004	5.03
	Higher	−0.028	0.102	−0.003	3.61
Working	No®				
	Yes	−0.010	−0.079	0.001	−1.0219
Religion	Hindu®				
	Muslim	−0.001	0.017	0.000	0.02
	Others	−0.004	0.037	0.000	0.17
Residence	Urban®				
	Rural	0.074	−0.509	−0.038	47.04
State	North®				
	Central	−0.018	−0.108	0.002	−2.42
	East	0.038	−0.294	−0.011	13.88
	North-east	0.005	−0.043	0.000	0.27
	West	0.015	0.109	0.002	−2.09
	South	0.027	0.175	0.005	−5.96
Mass media exposure	No®				
	Yes	−0.016	0.453	−0.007	9.15
Sex of household head	Male®				
	Female	0.031	0.042	0.001	−1.66
Age of household head	< 30 years®				
	30–44 years	0.008	−0.033	0.000	0.31
	45–59 years	−0.010	−0.026	0.000	−0.33
	60 and above	−0.018	0.038	−0.001	0.87
Tried to delay pregnancy	No®				
	Yes	−0.428	0.047	−0.020	25.05
Knowledge of any method	No®				
	Yes	−0.006	−0.015	0.000	−0.12
Family planning mass media exposure	No exposure®				
	One mass media	0.046	−0.229	−0.011	13.29
	Two mass media	0.025	0.015	0.000	−0.47
	Three mass media	0.008	0.169	0.001	−1.72
Knowledge of ovulatory cycle	Incorrect®				
	Correct	0.063	−0.047	−0.003	3.72
	Don't know	0.018	0.028	0.001	−0.65
				−0.080	

®Reference Category

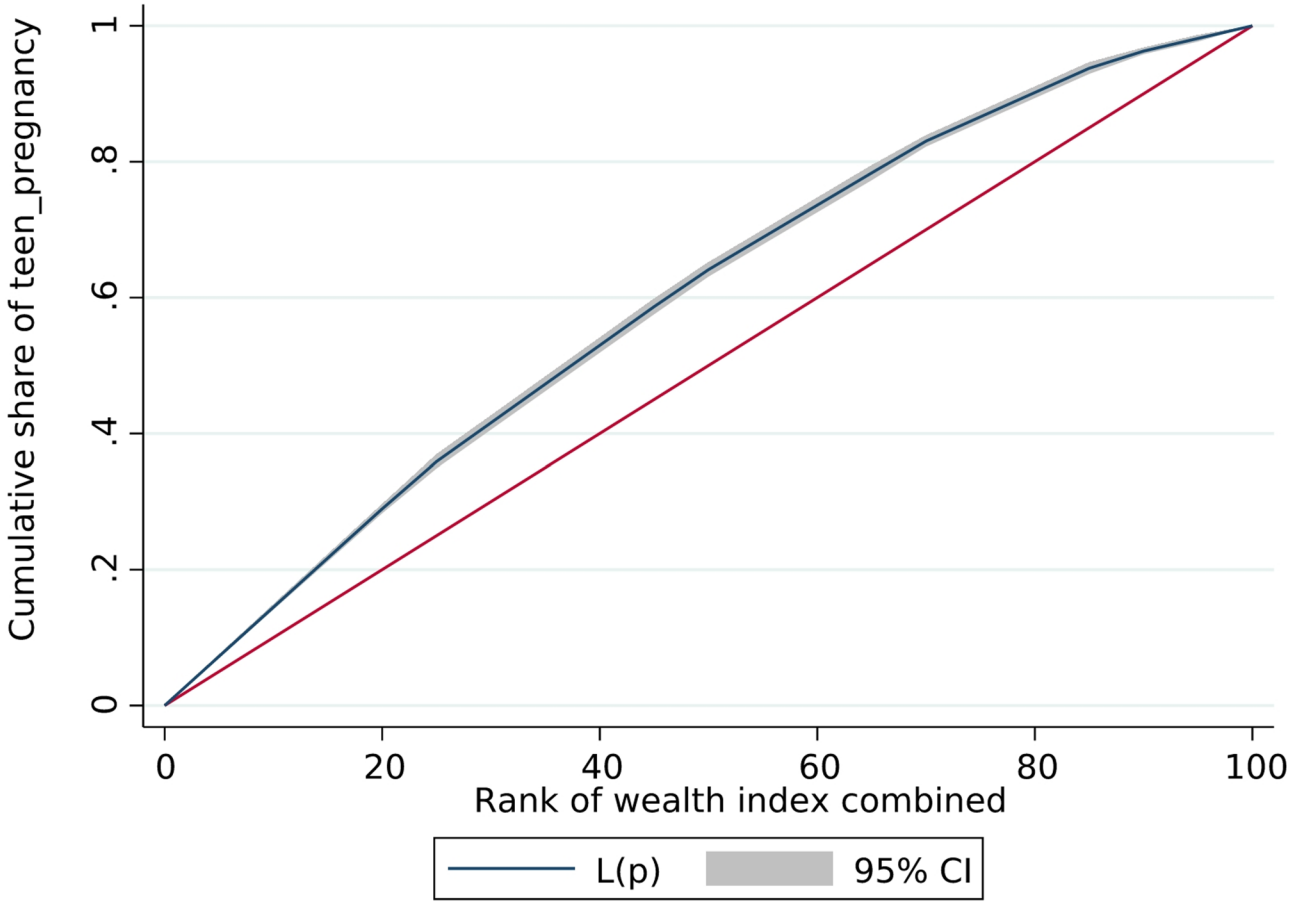


Fig. 1 Concentration curve for teenage pregnancy in India, from Erregyers decomposition

Discussion

In this study, we aimed to explore the factors that put young women at risk of pregnancy in India. Since the rates of teenage pregnancy when translated into absolute numbers are quite high in the country, understanding the predictors of teenage pregnancy is crucial from a policy and developmental perspective. Findings suggest that low educational attainment, poor wealth quintile, not working status, rural residence, and no mass media exposure shaped the pregnancy of teenage girls in India.

In line with our findings, several studies have reported higher pregnancy risk among girls without formal education or lower levels of education and those from poor wealth quintile [12–15]. Educated women likely have better knowledge about sexual and reproductive rights, including fertility and conception than those not educated [16]. Being enrolled in a school has been found to act as a protective factor against teenage pregnancy [17]. Adolescents who are not in school are denied access to comprehensive sexuality education as well as the skills necessary to negotiate sexuality and reproductive options and avoid pregnancy. This might also explain why uneducated girls are at higher risk of adolescent pregnancy. We further observed that the odds of teen pregnancy were 33% times lower among those who reported working. Education and employment are likely to increase autonomy and empowerment, and when combined with information about fertility management, allow young women to consider futures other than childbirth and unskilled labour. Empowering girls and women through formal education can certainly help address the higher rates of early marriage and teenage pregnancies [18] and break the intergenerational cycle of poverty [19].

Additionally, findings show that teenage pregnancy was mostly concentrated among rural residents. Past research has reported that girls residing in rural areas are mostly married off earlier than their urban counterparts due to economic, social, cultural, and religious factors [19]. Even young girls in rural regions are forced to marry because their parents believe marriage would protect them from sexual assault [20]. Surprisingly, it was found that half of the women who tried to delay/avoid pregnancy became pregnant. One of the probable reasons could be the exercise of their autonomy and their sexual and reproductive rights. Another annotation could be the proof of fertility as given by conception within a year in many cultural

traditions. Additionally, in a patriarchal society like India, the decision to have a child is not just a woman's decision but it is also determined by her husband, parents-in-law, and other relatives as well. These factors are likely to be interconnected, indicating that young women living in socioeconomically disadvantaged circumstances, with limited access to education or income-generating work, and who lack autonomy in deciding reproductive health or access to family planning, are at the highest risk of pregnancy during adolescence.

Findings from decomposition analysis indicated that factors such as rural residence (47%), tried to delay pregnancy, and no exposure to mass media related to family planning were the major contributors to observed wealth-based inequality. These findings are in line with the studies conducted in Nigeria which reported that wealth status, use of TV and the internet were significant predictors contributing to the concentration of teenage pregnancy among poor households [14]. The lower risk of teenage pregnancy among individuals who heard of family planning through TV, radio, or magazine/newspaper highlights the need for inclusion and reinforcement of family planning campaigns to facilitate the attainment of sustainable development goals by 2030 [19]. Additionally, measures and campaigns aimed at delaying the age of marriage can be beneficial in eliminating child marriages.

The higher risk of teenage pregnancy among those with incorrect knowledge of the ovulatory cycle and modern contraception methods is indicative of low self-efficacy in obtaining contraception and the lack of skills to negotiate their usage in preventing unwanted pregnancies as observed in Ghana [21] and other middle-income countries [22]. Moreover, barriers such as shyness in buying contraceptives, the cost of contraceptives and no-youth-friendly services also hinder the usage of contraception [23, 24]. Indeed, emphasizing communication with friends and partners about pregnancy risks and condom usage may be a significant extra component of treatments aiming at boosting self-efficacy and condom use skills among young women [21]. When categorised by age, older respondents i.e.18 or 19 years were more likely to be pregnant than their younger ones which was in line with the previous study [25]. However, to enhance the understanding, it is important to assess if the pregnancy was desired or unwanted. Further, the higher risk of pregnancy among those who tried to delay pregnancy was observed and the average marginal change in the risk of teenage pregnancy increased by 18% & 3% points when they tried to delay pregnancy and had knowledge of modern contraception respectively. It means that providing proper information about sexual behaviours and contraception to teens is critical. In addition, the prevalence of unmet need of contraception is quite high in the country (17.8% among adolescent women) [10] which might lead to unwanted pregnancies in the country.

Since birth occurring out of the marital union is nearly non-existent, the majority of these adolescent pregnancies are from the marital union in the country. However, several policies and programs addressing the age of marriage, have made substantial contributions in increasing the age at marriage [26] and consequently postponing teenage pregnancy to later ages. Given the high frequency of adolescent pregnancy in the country as a result of poor education and poverty, a focused formal education intervention is necessary. Healthcare practitioners should be prepared to give accurate, balanced sex education, including contraceptive and condom information, so that young people have the resources they need to protect themselves, all within the framework of healthy sexuality, free of stigma and judgment. Additionally, youngsters should be made aware of provisions of legal abortion and adequate post-abortion care services should be made available to them. The findings have important implications for achieving SDG 3 and SDG 5 targets. The observed socioeconomic disparities in teenage pregnancy highlight the urgent need for policies and programs that improve access to education, reproductive health information, and services among disadvantaged adolescents. Interventions aimed at empowering young girls, reducing early pregnancies, and promoting reproductive rights are critical steps towards gender equality and ensuring better health outcomes for adolescents.

Conclusions

To conclude, low educational attainment, poor wealth quintile, not working status, rural residence, and no mass media exposure shaped the pregnancy of teenage girls in India. Exposure to mass media related to family planning significantly reduced teenage pregnancy, thus awareness and knowledge regarding sexual and reproductive health, availability of contraception and their effective usage should be promoted. Future intervention strategies for adolescent pregnancy prevention must address the identified risk factors. Moreover, considering the disparity in the prevalence of teenage pregnancy, the implementation of existing programs and policies should be made more even. Promoting formal, institutional sources of knowledge e.g. School education can enable adolescents to understand the negative repercussions of adolescent pregnancy. Additionally, work participation, higher schooling and access to family planning can also positively contribute to the reduction of teenage pregnancies.

Strengths and limitations

The main strength of this paper is the use of nationally representative data and the application of Erreygers decomposition to analyse the factors predicting the

observed socio-economic inequality in teenage pregnancy in India. Since the study is cross-sectional, only associations can be established. Further, since the survey data was self-reported, the issue of recall bias and social desirability may occur. Another limitation of this study is that the analysis is restricted to adolescents aged 15 to 19 years. As a result, some respondents who had not yet experienced a pregnancy at the time of the survey may still be at risk of becoming pregnant before the end of adolescence. This could lead to an underestimation of the true burden of adolescent pregnancy.

Future Research Direction

Given the cross-sectional nature of the data used in this study, future research should focus on longitudinal studies that follow adolescents over time to better capture the dynamic risk of teenage pregnancy throughout adolescence. Further investigations are needed to explore the causal pathways linking socioeconomic inequalities, reproductive health knowledge, and teenage pregnancy outcomes. In particular, qualitative research could provide deeper insights into how self-efficacy, cultural norms and access to reproductive health services influence adolescents' pregnancy risk. Additionally, future studies should examine the effectiveness of targeted interventions such as comprehensive sexuality education and empowerment programs, particularly among marginalized and rural populations, to design evidence-based strategies for reducing teenage pregnancy and achieving SDG targets.

Abbreviations

SDG	Sustainable Development Goal
NFHS	National Family Health Survey
CI	Concentration Index
SCs	Scheduled Castes
STs'	Scheduled Tribes
OBCs	Other Backward Castes

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Authors' contributions

S.S. and N.S. conceptualized, analyzed the data, wrote the main manuscript text and prepared figures. Both authors reviewed the manuscript.

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Data availability

The data used in this study is available publicly for use at the website dhsprogram.com.

Declarations

Ethical approval and consent to participate

The analysis is based on secondary data available in the public domain for research; thus, no approval was required from any institutional ethical review board.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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